



**Generation Interconnection
System Impact Study Report
for
Queue Project AE1-059
STANTON-SUMMIT 230 KV
250 MW Capacity / 500 MW Energy**

August, 2019

1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

2 General

The IC has proposed a hydro (pumped storage) generating facility located in Luzerne County, Pennsylvania. The installed facilities will have a total capability of 500 MW with 250 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project is December 15, 2022. **This study does not imply a PPL EU commitment to this in-service date.**

Queue Number	AE1-059
Project Name	STANTON-SUMMIT 230 KV
Interconnection Customer	Renewable Energy Aggregators
State	Pennsylvania
County	Luzerne
Transmission Owner	PPL
MFO	500
MWE	500
MWC	250
Fuel	Hydro
Basecase Study Year	2022

2.1 Point of Interconnection

AE1-059 will interconnect with the PPL EU transmission system to the Stanton – Summit #4 230kV line via a new PPL EU 230kV switchyard, 1.5 miles from the Stanton 230/69kV substation and 7.56 miles from the Summit 230/69kV substation. The POI will be where the IC Attachment Facilities (with insulators) connects to the first dead-end structure inside the new PPL EU switchyard.

The scope of work provided in this System Impact Study is for the facilities to be constructed on the PPL EU side of the POI.

2.2 Cost Summary

The AE1-059 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$19,147,000
Non Direct Connection Network Upgrades	\$2,272,872
Total Costs	\$21,419,872

In addition, the AE1-059 project will be responsible for the following allocated costs for System Upgrades:

Description	Total Cost
System Upgrades	\$37,350,000*

* Stability Analysis and NYISO impact evaluation will be performed during the Facilities Study. **Note that additional reinforcements may be identified to mitigate stability concerns.**

3 Transmission Owner Scope of Work

3.1 Attachment Facilities

None. It is assumed the IC will construct the estimated 2.75 miles of generator lead line from the Customer Facility to the Point of Interconnection - the first dead-end structure inside the new interconnection switchyard.

3.2 Direct Connection Cost Estimate

New Switchyard

- Construct one new standard two bay (Breaker and a Half arrangement) 230kV switchyard at the GPS Coordinates 41.23.31.65N, -75.46.51.08W interconnected to the Stanton – Summit #4 230kV transmission line.
- The 230kV Stanton line and one new 230kV IC generator lead line will terminate into one bay.
- The 230 kV Summit line and one new 230kV IC generator lead line will terminate into the other bay.

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
New Switchyard	\$19,147,000
Total Direct Connection Costs	\$19,147,000

3.3 Non-Direct Connection Cost Estimate

Transmission Line Tap

- Tap the existing Stanton-Summit #4 230kV line.
- Install two dead-end tap poles and route the lines into the new 230kV switchyard.
- Install conductor using 1590 ACSR and dual OPGW.
- Install tangent tension structure on the Stanton-Summit switchyard location assumed to be adjacent to the 230kv corridor.

Remote End Work

Develop all engineering packages including BOM, construction support and close-out in compliance with PPL EU Standards and Specs:

- Model IC in CAPE and conduct a wide area short-circuit study at least two busses away from the IC facilities. Identify affected relays and revise settings as needed.

- Revise protective relay settings at both Stanton and Summit Substations to evaluate the effects of infeed from the IC.
- Revise DTT schemes at both Stanton and Summit Substations to reflect new IC switchyard.
- At Stanton Substation, update all Summit line designations on equipment, panels, and drawings. Items associated with this line should be updated to reference the new IC Switchyard.
- At Summit Substation, update all Stanton line designations on equipment, panels, and drawings. Items associated with this line should be updated to reference the new IC Switchyard.

Transmission Fiber

- Break the existing OPGW on the Shickshinny – Lackawanna 500kV line, near the proposed new switchyard location
 - Install two new splice boxes
 - Install new span of OPGW
- Break the existing OPGW on the Shickshinny - Lackawanna 500kV line at a second point, near Summit substation
 - Install two new splice boxes
 - Install new span of OPGW
- Install OPGW to both the new 230kV Stanton tap pole and the new switchyard dead-end.
 - Run fiber from dead-end to switchyard control house
- Install OPGW to both the new 230kV Summit tap pole and the new switchyard dead-end.
 - Run fiber from dead-end to switchyard control house

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Transmission Line Tap	\$1,472,455
Remote End Work – Stanton Substation	\$108,000
Remote End Work – Summit Substation	\$108,000
Transmission Fiber	\$584,417
Total Non-Direct Connection Facility Costs	\$2,272,872

3.4 Estimated Schedule

The estimated time to complete the Transmission Owner scope of work is **24-36 months** after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

3.5 Assumptions

- Availability of optimal transmission line route.

- Outage feasibility not assessed until Facilities Study.
- No major environmental, real estate, or permitting issues.
- IC is responsible for acquisition of easements and rights of way for the Attachment Facilities.

4 Interconnection Customer Requirements

4.1 PPL Requirements

PPL EU Interconnection Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

IC Substation POC Fault Interrupting Device (FID) Requirements

Based on the latest conceptual single line diagram provided by the IC, no POC FIDs have been provided. Per PPL EU Interconnection Requirements found at the link above, two (2) 230 kV rated circuit breakers in this case, shall be installed and equipped with dual trip coils and capable of interrupting worst-case scenario fault currents with a rated speed of two (2) cycles or less. The POC FID circuit breakers shall be operated by their respective primary and backup and DTT relaying equipment.

5 Revenue Metering and SCADA Requirements

5.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

5.2 PPL Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

6 Network Impacts

The Queue Project AE1-059 was evaluated as a 500.0 MW (Capacity 250.0 MW) injection at Stanton-Summit 230kV in the PPL area. Project AE1-059 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-059 was studied with a commercial probability of 1.00. Potential network impacts were as follows:

Summer Peak Load Flow

7 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

8 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3351399	208009	LACK	PPL	200074	LACKAW	PJM	3	PL:02:P42:001057	breaker	1165.0	89.41	103.81	AC	173.78
3351654	208040	MONT	PPL	208034	MILT	PPL	1	PL:08:P71:101343	tower	739.0	97.28	100.21	AC	26.16

9 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
4103863	200675	26E.TWANDA	PENELEC	130763	HILSD230	NYISO	1	PL:03:P42:000127	breaker	531.0	136.03	140.55	DC	27.67
4103864	200675	26E.TWANDA	PENELEC	130763	HILSD230	NYISO	1	PL:02:P42:100875	breaker	531.0	135.86	140.38	DC	27.67
4104330	200675	26E.TWANDA	PENELEC	130763	HILSD230	NYISO	1	PS_P1-2_5063	single	531.0	134.96	136.72	AC	12.77
4104675	200675	26E.TWANDA	PENELEC	130763	HILSD230	NYISO	1	PL:03:P71:001017	tower	531.0	140.2	144.84	DC	28.43

10 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
4104508	200021	SUNBURY	PJM	200009	JUNIATA	PJM	1	PL:28:P12:000080	operation	3112.0	100.98	105.78	AC	174.51
4104589	200022	SUSQHANA	PJM	200023	WESCOVLE	PJM	1	PL:08:P12:000083	operation	3112.0	95.13	99.29	AC	148.2
4104586	200023	WESCOVLE	PJM	200075	BREI	PJM	1	PL:08:P12:000083	operation	3112.0	95.46	99.92	AC	159.27
4104326	200675	26E.TWANDA	PENELEC	130763	HILSD230	NYISO	1	PS_P1-2_5063	operation	531.0	135.12	139.3	DC	25.54

11 Stability and Reactive Power Requirement for Low Voltage Ride Through

Stability Analysis will be performed during the Facilities Study. **Note that additional reinforcements may be identified to mitigate stability concerns.**

12 Light Load Analysis

No mitigations are required.

13 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-059	NUN												
3351399	1	LACK T3 500/230 kV	Re-Terminate LACK AWANNA 500/230kV T4 transformer	\$4,000,000	\$4,000,000	N6167												
3351654	2	MONT-MILT 230 kV	Build a second circuit from MONTOUR -MILT ON 230kV and MILTON-SUNBURY 230kV	\$33,000,000	\$33,000,000	N6168												
4103863, 4103864, 4104330, 4104675	3	East Towanda - Hillside (NYSEG) 230 kV Line	The overloaded Facility is a Tie line with a portion owned by PENELEC and the balance owned by NYSEG. <u>PENELEC Upgrade:</u> Re-conductor the Penelec portion of the East Towanda – Hillside 230 kV line and upgrade terminal equipment at East Towanda substation. Cost: \$ 60,259,000 Time: 27 Months Rating after the upgrade: 855/984/984 <u>***Note 1:</u> Although Queue Project AE1-059 may not have cost responsibility for this upgrade, Queue Project AE1-059 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-059 comes into service prior to completion of the upgrade, Queue Project AE1-059 will need an interim study. The potential cost responsibility is as follows: <table><tr><th>Queue</th><th>MW contribution</th><th>% Cost Allocation</th><th>Cost (\$ 60,259,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-059</td><td>28.43</td><td>100.00%</td><td>\$60,259,000</td><td>PL-03:P71:001 017</td><td>tower</td></tr></table>	Queue	MW contribution	% Cost Allocation	Cost (\$ 60,259,000)	Contingency Name	Contingency Type	AE1-059	28.43	100.00%	\$60,259,000	PL-03:P71:001 017	tower	\$60,259,000	\$TBD***	N6169
			Queue	MW contribution	% Cost Allocation	Cost (\$ 60,259,000)	Contingency Name	Contingency Type										
AE1-059	28.43	100.00%	\$60,259,000	PL-03:P71:001 017	tower													
<u>NYISO Upgrade:</u> Replace Wave trap at Hillside Substation Re-conductor overhead wire on the NYSEG portion of the line with 1033.5 ACSR cable Re-conductor overhead wire on the NYSEG portion of the line with 1033.5 ACSR cable Cost: \$40,634,474 Rating after the upgrade: 669/809/940 IC will need to enter into an Affected System Study Agreement with NYISO to evaluate upgrade cost and allocation. <u>**Note 2:</u> Although Queue Project AE1-059 may not have cost responsibility for this upgrade, Queue Project AE1-059 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-059 comes into service prior to completion of the upgrade, Queue Project AE1-059 will need an interim study.	\$40,634,474	\$TBD**	N6169															
Not Applicable (N/A)	NA	230kV Breaker	Replace breaker on the Clinton –Sunbury 230kV line Estimated Cost: \$350,000 (See Short Circuit section of this report for details of this overload)	\$350,000	\$350,000	Pending												
			TOTAL COST	\$138,243,474	\$37,350,000*													

* Estimated total cost assumes that cost allocation for the East Towanda – Hillside 230 kV overload mitigation is \$0.

14 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

14.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3351399	208009	LACK	PPL	200074	LACKAW	PJM	3	PL:02:P42:001057	breaker	1165.0	89.41	103.81	AC	173.78

Bus #	Bus	MW Impact
200823	26MHP_X3-003	17.38
200894	26K02	9.8
203999	P-047 E	14.14
208055	LAEN	87.32
209009	PEIP 2	2.79
210706	HOLD	9.02
294573	P-028 E	35.17
916202	Z1-069 E	6.93
917661	WAYM E	25.2
918601	AA1-077 C	2.17
918602	AA1-077 E	35.81
918682	AA1-082 E	13.51
919201	AA1-144 O1	37.6
919491	AA2-000	52.7
920341	AA2-132	4.3
930511	AB1-092	1.93
931092	AB1-160 E	1.98
931942	AB1-182 E	4.7
932571	AC2-077	2.48
936421	AD2-055	3.98
938331	AE1-051	4.53
938401	AE1-059 C O1	86.89
938402	AE1-059 E O1	86.89
CBM-N	CBM-N	0.34
CBM-S1	CBM-S1	0.79
CBM-S2	CBM-S2	0.04
CBM-W1	CBM-W1	2.35
CBM-W2	CBM-W2	6.63
CIN	CIN	0.85
G-007	G-007	4.66
HAMLET	HAMLET	0.01
IPL	IPL	0.55
LGEE	LGEE	0.23
MEC	MEC	1.54
MECS	MECS	1.84
NYISO	NYISO	1.24
O-066	O-066	32.36
WEC	WEC	0.24

14.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3351654	208040	MONT	PPL	208034	MILT	PPL	1	PL:08:P71:101343	tower	739.0	97.28	100.21	AC	26.16

Bus #	Bus	MW Impact
208911	MONT G1	27.95
208912	MONT G2	28.31
209006	NEPC IPP	3.63
211375	BEAC	1.4
212369	PATRIOT 1	14.07
212370	PATRIOT 2	14.07
292935	U2-015 E	6.4
919512	AA2-008 E	15.18
926681	AC1-151 C	0.53
926682	AC1-151 E	0.87
931942	AB1-182 E	1.29
938401	AE1-059 C O1	13.08
938402	AE1-059 E O1	13.08
939521	AE1-181 C	1.89
939522	AE1-181 E	1.26
BLUEG	BLUEG	3.57
CALDERWOOD	CALDERWOOD	0.38
CANNELTON	CANNELTON	0.22
CATAWBA	CATAWBA	0.24
CBM-N	CBM-N	0.67
CHEOAH	CHEOAH	0.35
CHILHOWEE	CHILHOWEE	0.12
COFFEEN	COFFEEN	0.38
COTTONWOOD	COTTONWOOD	1.47
DUCKCREEK	DUCKCREEK	0.82
EDWARDS	EDWARDS	0.37
ELMERSMITH	ELMERSMITH	0.37
FARMERCITY	FARMERCITY	0.25
G-007A	G-007A	1.67
GIBSON	GIBSON	0.15
HAMLET	HAMLET	0.39
NEWTON	NEWTON	0.98
NYISO	NYISO	2.89
PRAIRIE	PRAIRIE	1.83
SANTEETLA	SANTEETLA	0.1
SMITHLAND	SMITHLAND	0.15
TATANKA	TATANKA	0.45
TILTON	TILTON	0.45
TRIMBLE	TRIMBLE	0.4
TVA	TVA	1.23
UNIONPOWER	UNIONPOWER	0.55

Bus #	Bus	MW Impact
VFT	VFT	5.44

14.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
4104675	200675	26E.TWANDA	PENELEC	130763	HILSD230	NYISO	1	PL:03:P71:001017	tower	531.0	140.2	144.84	DC	28.43

Bus #	Bus	MW Impact
200823	26MHP_X3-003	7.65
200851	26MEHOOP3	1.07
200894	26K02	9.26
200917	26MTNTP_P28	0.54
200949	26X1-109	26.72
203283	26MANOR_T86	0.03
210706	HOLD	1.27
294573	P-028 E	15.48
913191	Y1-047 OP1	0.36
916541	Z1-110	0.37
917072	Z2-011	0.37
917661	WAYM E	3.56
918602	AA1-077 E	5.06
918682	AA1-082 E	14.28
919201	AA1-144 O1	39.49
919491	AA2-000	40.1
920351	AA2-133	0.41
925951	AC1-071 C	0.55
925952	AC1-071 E	3.71
930511	AB1-092	1.47
934801	AD1-108	0.03
934811	AD1-109	0.02
934821	AD1-110	0.03
936421	AD2-055	3.03
938331	AE1-051	0.64
938401	AE1-059 C O1	14.21
938402	AE1-059 E O1	14.21
AA3-200	AA3-200	47.57
AA3-300	AA3-300	7.92
AA3-700	AA3-700	11.33
BLUEG	BLUEG	4.36
CALDERWOOD	CALDERWOOD	0.41
CANNELTON	CANNELTON	0.26
CARR	CARR	2.27
CATAWBA	CATAWBA	0.23
CHEOAH	CHEOAH	0.37
CHILHOWEE	CHILHOWEE	0.13
COFFEEN	COFFEEN	0.46
COTTONWOOD	COTTONWOOD	1.67
DUCKCREEK	DUCKCREEK	1.02
EDWARDS	EDWARDS	0.47
FARMERCITY	FARMERCITY	0.3
G-007A	G-007A	2.44

Bus #	Bus	MW Impact
GIBSON	GIBSON	0.18
HAMLET	HAMLET	0.36
NEWTON	NEWTON	1.2
PRAIRIE	PRAIRIE	2.2
RENSSELAER	RENSSELAER	1.82
SANTEETLA	SANTEETLA	0.11
SMITHLAND	SMITHLAND	0.17
TATANKA	TATANKA	0.55
TILTON	TILTON	0.56
TRIMBLE	TRIMBLE	0.48
TVA	TVA	1.38
UNIONPOWER	UNIONPOWER	0.6
VFT	VFT	6.46

Affected Systems

15 Affected Systems

15.1 NYISO

A NYISO Affected System Study will be required for AE1-059. The AE1-059 IC will need to enter into an Affected System Study Agreement with New York State Electric and Gas Corporation.

Contingency Name	Contingency Definition
PL:08:P71:101343	CONTINGENCY 'PL:08:P71:101343' /* SAEGERS-CLINTON & SAEGERS-ELIMSPORT 230 KV TOWER DISCONNECT BRANCH FROM BUS 212397 TO BUS 207968 CKT 1 /* /* SAEG-ELIM DISCONNECT BRANCH FROM BUS 207937 TO BUS 212397 CKT 1 /* /* SAEG-CLIN END
PL:03:P71:001017	CONTINGENCY 'PL:03:P71:001017' /* LACK-HOPA 500KV & PAUP-BLGR 230KV DISCONNECT BRANCH FROM BUS 200074 TO BUS 200098 CKT 1 /* LACKAW-LACHOP_T 500 DISCONNECT BRANCH FROM BUS 200091 TO BUS 200098 CKT 1 /* HOPAKONG-LACHOP_T 500 DISCONNECT BRANCH FROM BUS 208049 TO BUS 207919 CKT 1 /* PAUP-BLGR TR2 230 DISCONNECT BRANCH FROM BUS 207919 TO BUS 211394 CKT 2 /* BLGR-BLGR 230/69KV T2 END
PS_P1-2_5063	CONTINGENCY 'PS_P1-2_5063' DISCONNECT BUS 200098 /* LACKAWANNA HOPATCONG_T END
PL:08:P12:000083	CONTINGENCY 'PL:08:P12:000083' /* JUNI-SUNB 500KV LINE DISCONNECT BRANCH FROM BUS 200009 TO BUS 200021 CKT 1 /* JUNIATA-SUNBURY 500 END
PL:28:P12:000080	CONTINGENCY 'PL:28:P12:000080' /* SUSQ-WESC 500KV LINE DISCONNECT BRANCH FROM BUS 200022 TO BUS 200023 CKT 1 /* SUSQHANA-WESCOVLE 500 END
PL:02:P42:001057	CONTINGENCY 'PL:02:P42:001057' /* LACK-PAUP 230KV; PAUP_E CB @ LACK DISCONNECT BRANCH FROM BUS 208009 TO BUS 925950 CKT 1 /* LACK-AC1-071 TAP 230 /* CONTINGENCY LINE ADDED FOR AE1 BUILD DISCONNECT BRANCH FROM BUS 208009 TO BUS 211681 CKT 2 /* LACK-LACK 230/69KV T2 DISCONNECT BRANCH FROM BUS 208009 TO BUS 200074 CKT 4 /* LACK-LACKAW 500/230KV T4 END
PL:02:P42:100875	CONTINGENCY 'PL:02:P42:100875' /* LACK-HOPA 500KV; HOPA_E CB @ LACK DISCONNECT BRANCH FROM BUS 200098 TO BUS 200074 CKT 1 /* DISCONNECT BRANCH FROM BUS 200091 TO BUS 200098 CKT 1 /* DISCONNECT BRANCH FROM BUS 200074 TO BUS 208009 CKT 4 /* LACKAW-LACK 500-230 T4 END
PL:03:P42:000127	CONTINGENCY 'PL:03:P42:000127' /* LACK-HOPA 500KV; HOPA_E CB @ LACK 500KV DECREASE BUS 200074 SHUNT BY 100 PERCENT /* LACK 500KV REACTOR DISCONNECT BRANCH FROM BUS 200091 TO BUS 200098 CKT 1 /* HOPAKONG-LACJEF T 500 DISCONNECT BRANCH FROM BUS 200074 TO BUS 208009 CKT 4 /* LACKAW-LACK 500-230 DISCONNECT BRANCH FROM BUS 200074 TO BUS 200098 CKT 1 /* LACKAW-LACJEF T 500 END
PL:1A:P42:000923	CONTINGENCY 'PL:1A:P42:000923' /* SUNBURY 500KV YARD 3T BF DISCONNECT BRANCH FROM BUS 200021 TO BUS 200009 CKT 1 /* /* JUNIATA-SUNBURY 500KV LINE DISCONNECT BRANCH FROM BUS 200021 TO BUS 208109 CKT 25 /* /* T25 END

Contingency Name	Contingency Definition
PL:1A:P42:000922	CONTINGENCY 'PL:1A:P42:000922' /* SUNBURY 500KV YARD 3N BF DISCONNECT BRANCH FROM BUS 200021 TO BUS 200009 CKT 1 /* /* JUNIATA-SUNBURY 500KV LINE DISCONNECT BRANCH FROM BUS 200021 TO BUS 208109 CKT 24 /* /* T24 END

Short Circuit

16 Short Circuit

The following Breakers are over duty

Bus Number	Bus Name	BREAKER	Type	Capacity (Amps)	Duty Percentage Post Queue	Duty Percentage Pre Queue
208040	Mont 230kV	CLINTON-SUNB	S	50000	103.78%	98.62%

Please refer to System Reinforcements section 13 above for cost to mitigate the Short Circuit violation

Attachment 1 – Single Line Diagram

